

Q1 2021 (31 Aug Shift 1)

A body of mass M moving at speed V_0 collides elastically with a mass ' m ' at rest. After the collision, the two masses move at angles θ_1 and θ_2 with respect to the initial direction of motion of the body of mass M . The largest possible value of the ratio M/m , for which the angles θ_1 and θ_2 will be equal, is :

(1) 4

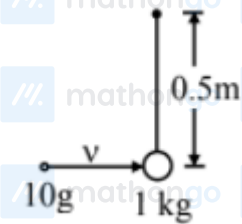
(2) 1

(3) 3

(4) 2

Q2 2021 (27 Aug Shift 2)

A bullet of 10 g, moving with velocity v , collides head-on with the stationary bob of a pendulum and recoils with velocity 100 m/s. The length of the pendulum is 0.5 m and mass of the bob is 1 kg. The minimum value of $v = m/s$ so that the pendulum describes a circle. (Assume the string to be inextensible and $g = 10 \text{ m/s}^2$)

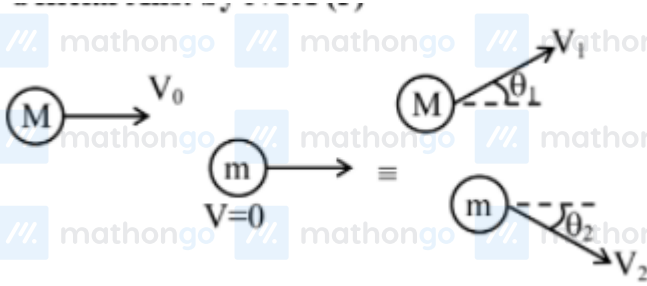


Answer Key

Q1 (3)

Q2 (400)

Q1 (3)

given $\theta_1 = \theta_2 = \theta$

from momentum conservation

in x -direction $MV_0 = MV_1 \cos \theta + mV_2 \cos \theta$ in y -direction $0 = MV_1 \sin \theta - mV_2 \sin \theta$

Solving above equations

$$V_2 = \frac{MV_1}{m}, \quad V_0 = 2V_1 \cos \theta$$

From energy conservation

$$\frac{1}{2}MV_0^2 = \frac{1}{2}MV_1^2 + \frac{1}{2}mV_2^2$$

Substituting value of V_2 & V_0 , we will get

$$\frac{M}{m} + 1 = 4 \cos^2 \theta \leq 4$$

$$\frac{M}{m} \leq 3$$

Option (3)

Q2 (400)



$$V' = \sqrt{5gR} = \sqrt{5 \times 10 \times 0.5}$$

$$V' = 5 \text{ m/s}$$

$$m_1 V = m_2 \times 5 - m_1 \times 100$$

$$\frac{10}{1000} \times V = 5 - \frac{10}{1000} \times 100$$

$$V = 400 \text{ m/s}$$